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The Veterans Affairs Low-Vision Visual Functioning Questionnaire-48 (VA LV VFQ-48): Performance of the Italian version

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Abstract

Purpose: The Veterans Affairs Low-Vision Visual Functioning Questionnaire-48 is among the most validated tools to collect patient-reported outcomes in a low-vision population. We have aimed to conduct a pilot validation of the Italian version of the Veterans Affairs Low-Vision Visual Functioning Questionnaire-48.

Methods: The Veterans Affairs Low-Vision Visual Functioning Questionnaire-48 was translated using a standardized procedure and then administered to consecutive low-vision patients attending rehabilitation services in three centers. Patients were interviewed by a trained psychologist regarding the individual items of the tool.

Results: We included 131 patients with a mean visual acuity of 0.91 logMAR (standard deviation: 0.42 logMAR), mostly affected by age-related macular degeneration. The Veterans Affairs Low-Vision Visual Functioning Questionnaire-48 showed high internal consistency (Cronbach's alpha: 0.98) and good item-test and item-rest correlation (median: 0.73 and 0.71, respectively). Both the overall score and the subscale (reading, visual motor, mobility and visual information) scores significantly correlated with visual acuity, reading acuity and speed. Reading speed achieved the best absolute correlation with the Veterans Affairs Low-Vision Visual Functioning Questionnaire-48 scores (Spearman r: 0.39–0.49).

Conclusion: The Italian version of the Veterans Affairs Low-Vision Visual Functioning Questionnaire-48 is a valid tool to assess patients attending low-vision services. Revising a few items may further improve the tool.

Keywords

Low-vision, rehabilitation, quality of life, activities of daily living, questionnaires

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Introduction

The aim of low vision rehabilitation (LVR) is to improve the quality of life (QOL) of incurable visually impaired people. The assessment of the needs and abilities of these patients, such as the outcomes of LVR, must be patient-centered and measurable. Therefore, a tool for good patient-centered outcome (PCO) measures is necessary in order to assess and compare the QOL of each patient during LVR.

A direct measure can be obtained during observation of a patient attempting a specific task, usually scored on the time required and/or on the number of errors made, but recently many authors have preferred a patient-reported outcome (PRO) measure to assess the ability to complete the same task.[1] Some questionnaires have been developed in past years with this aim, in particular five instruments studying vision-related functions in low vision patients.[2–6,7]

In particular, the Veterans Affairs Low-Vision Visual Functioning Questionnaire-48 (VA LV VFQ48)[8] is a 48-item questionnaire developed and validated by Stelmack to measure the PCO of low-vision patients undergoing LVR. The 48-item content reflects the activities that are targeted by LVR, which depend heavily on vision, and are important to most patients.[9] This instrument is able to measure LVR results[10] and was selected to be used in the "Low Vision Intervention Trial" (LOVIT and LOVIT II).[11,12]

The aim of this study is to develop an Italian version of the VA LV VFQ48 to be integrated into a web-based register of visually impaired patients undergoing LVR in many LVR centers in Italy.

Methods

Adult patients (age 18 years or more) were recruited at three low-vision services located in Padua, Florence, and Rome. There was no selection based on visual acuity or disease, since we aimed to recruit a representative sample of patients in need of vision rehabilitation, including central or peripheral visual loss.

The primary goal was to assess the usability and internal consistency of the translated version of the VA LV VFQ48, which was administered to all patients by trained psychologists. Visual acuity (Early Treatment Diabetic Retinopathy Study (ETDRS) charts), contrast sensitivity (Pelli–Robson charts) and reading performance data (MNREAD charts) were available for a subset of patients when an ophthalmic examination was conducted the same day as the psychological assessment.

The back-translation procedure was used to translate and adapt the VA LV VFQ-48 questionnaire to the Italian context, based on Vijver and Tanzer's suggestions.[13] Once the back-translation was complete, the VA LV VFW-48 was administered (the assessment phase was conducted in two one-to-one interviews which lasted approximately 60 min each), asking participants to respond and reformulate the items, indicate unclear words, and identify possible difficulties. The Italian version of the VA LV VFQ-48 (included in supplementary material) was used with other instruments to assess depression (Beck Depression Inventory),[14] life satisfaction (Satisfaction with Life Scale),[15] hope (The Hope Scale),[16] optimism (Life Orientation Test),[17] and courage (Courage scale).[18] These instruments were used to examine the validity of the Italian version of the VA LV VFQ-48, which will be the subject of a separate report, and to provide all participants with a personalized report on the results achieved.

We used descriptive statistics and histograms to report on the score and missing data structure of each item in the questionnaire. The internal consistency of the questionnaire was assessed using Cronbach's alpha. Item-test correlation and item-rest correlation were also calculated. Criterion validity was studied using visual function measures as an external criterion, including visual acuity (logMAR), contrast sensitivity (log10 units), reading acuity (logMAR), and reading speed (words/minute). For this purpose, we computed Spearman correlation between the overall score and each subscale domain score with visual function measures. In order to check the robustness of our correlation estimates using the averages of the raw questionnaire scores (1 to 5), we also extracted the person-specific latent trait using an Item Response Theory model for ordinal responses (Graded Response Model, GRM)[19] fitted by means of the *irt grm* command in Stata 15.1 (StataCorp, College Station, TX), which was used for all our statistical analyses. The study followed the tenets of the Declaration of Helsinki and was approved by the Local Institutional Review Board.

Results

Sample characteristics and questionnaire performance

For the validation of the Italian version of the VA-VFQ-48, we interviewed 131 patients who received the questionnaire and a battery of psychological tests and other questionnaires, with the purpose of confirming whether the translation met an Italian setting. This work was also based on a series of focus groups and will be the subject of a separate report. The mean age was 69 years (standard deviation (SD): 14 years) and 75 out of 131 patients (54%) were female. Age-Related Macular Degeneration (AMD) was the most common cause of visual impairment (58 subjects), followed by myopic macular degeneration (n = 27), diabetic retinopathy (n = 21), glaucoma (n = 14), and a miscellanea of acquired and inherited diseases (n = 21). In order to confirm the validity of the translated VFQ, using visual function and reading performance as external criteria, we computed Cronbach's alpha and found it very high and reliable (alpha: 0.98).

Table 1. Mean score for each item, together with any missing data and test/rest correlation.

Item	Description	Number responses	Mean score	SD	%Missing	Item-test correlation	item-rest correlation
1	Physically get dressed	116	1.10	0.41	11	0.25	0.24
2	Keep clean/keep your clothes clean	105	1.80	0.91	20	0.67	0.65
3	Identify medicine	127	1.70	0.83	3	0.60	0.58
4	Tell time	130	1.80	0.91	1	0.66	0.65
5	Identify money	130	2.10	0.79	1	0.64	0.62
6	Match clothes	126	1.60	0.90	4	0.67	0.65
7	Groom yourself	126	1.60	0.70	4	0.54	0.52
8	Identify food on plate	130	1.80	0.86	1	0.69	0.68
9	Eat and drink neatly	129	1.60	0.71	2	0.54	0.52
10	Fix a snack	120	1.40	0.71	8	0.67	0.66
11	Prepare meals	104	1.70	0.91	21	0.78	0.76
12	Use appliance dials	106	1.80	0.99	19	0.76	0.75
13	Clean the house	85	1.90	0.91	35	0.78	0.77
14	Handle finances	109	2.10	1.10	17	0.75	0.73
15	Make out a check	75	2.70	1.10	43	0.72	0.70
16	Take a message	118	2.00	0.97	10	0.75	0.74
17	Find something on crowded shelf	129	2.10	0.92	2	0.73	0.71
18	Find public restrooms	112	2.20	0.97	15	0.76	0.75
19	Get around indoors in places you know	127	1.30	0.61	3	0.55	0.53
20	Get around outdoors in places you know	121	1.70	0.89	8	0.75	0.74
21	Get around in unfamiliar places	119	2.70	0.97	9	0.81	0.80
22	Go out at night	73	3.00	1.10	44	0.72	0.70
23	Get down steps in dim light	114	2.30	0.90	13	0.70	0.68
24	Adjust to bright light	124	2.50	0.88	5	0.64	0.62
25	Cross streets at a traffic light	121	2.00	1.00	8	0.79	0.78
26	Use public transportation	89	2.60	1.10	32	0.73	0.71
27	Get around in a crowd	109	2.20	1.00	17	0.72	0.71
28	Avoid bumping into things	129	1.80	0.75	2	0.57	0.55
29	Recognize persons up close	129	1.90	0.97	2	0.69	0.67
30	Recognize persons across the room	129	2.80	1.10	2	0.77	0.75
31	Read street signs and store names	126	2.80	1.10	4	0.77	0.76
32	Read newspaper headlines	129	2.00	1.10	2	0.73	0.72
33	Read menus	121	2.90	1.10	8	0.78	0.77
34	Read newspapers/magazines	127	3.00	0.97	3	0.70	0.68
35	Read mail	127	2.80	1.10	3	0.79	0.78
36	Read small print on a package label	129	3.30	0.92	2	0.65	0.63
37	Read print on TV	128	3.10	1.00	2	0.73	0.71
38	Keep your place while reading	75	2.60	1.20	43	0.81	0.80
39	Watch TV	128	2.10	0.96	2	0.75	0.73
40	Play table and card games	96	2.60	1.20	27	0.80	0.79
41	See photos	127	2.40	1.10	3	0.79	0.77
42	Work on your favorite hobby	110	2.80	1.10	16	0.64	0.62
43	Go to the movies	57	2.60	1.10	56	0.79	0.77
44	Go to spectator events	75	2.30	1.10	43	0.80	0.78
45	Play sports	58	2.50	1.30	56	0.72	0.70
46	Do yard work	80	2.30	1.20	39	0.83	0.82
47	Sign your name	129	1.80	0.85	2	0.62	0.60
48	Read store signs	126	2.20	1.10	4	0.78	0.76

SD: standard deviation.

The median item-test correlation was 0.73 (interquartile interval: 0.67–0.77), with only one item (Item 1) having a correlation of less than 0.50 (0.26). Table 1 presents the mean score (SD) for each item, together with item-test and item-rest correlation. Figure 1 shows the distribution of the scores for each item. Ten items presented more than 25%

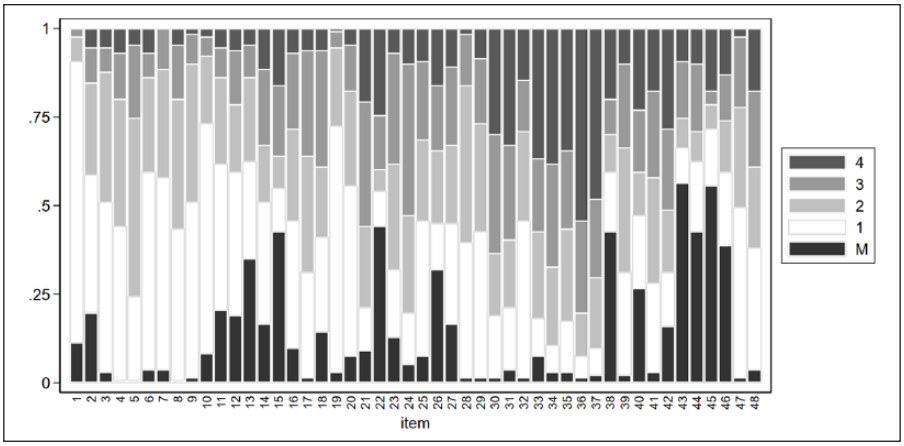


Figure 1. Distribution of the scores for each item, including missing data.

Table 2. Pairwise correlation (Spearman r) between functional variables and questionnaire overall as well as subscale scores.

	Overall score	Latent trait (Graded Response Model)	Reading subscale	Visual motor subscale	Mobility subscale	Visual information subscale
Visual acuity	0.34	0.36	0.37	0.31	0.24	0.38
Reading acuity	−0.37	−0.39	−0.30	−0.32	−0.34	−0.40
Contrast sensitivity	0.29	0.31	0.30	0.31	0.18	0.33
Reading speed	−0.44	0.48	−0.44	−0.39	−0.39	−0.49

missing responses, ranging from 27% for "Play table and card games" to 56% for "Go to movies." Considering the high frequency of omitted items and based on participants' indications, the following items request rephrasing: 1, 11, 13, 14, 15, 16, 22, 38, 40, 45, 46.

Association with visual and reading function data

Questionnaire data could be linked with best-corrected visual acuity (BCVA) data for 106 patients and with reading performance and contrast sensitivity data for 76 patients. There was minimal and non-significant Spearman correlation between age and overall or subscale scores (Spearman r between −0.02 and −0.13 in absolute values) and no association was found with sex in regression models.

Table 2 presents pairwise correlation (Spearman r) between the functional variables and the overall scores of the questionnaire, as well as the subscale scores. All coefficients were in the range of low to moderate correlation but statistically significant, except for the lowest value (reading acuity with mobility subscale score). Although we did not conduct formal comparisons of the associations between visual function measures and questionnaire scores, we observed that the highest correlation was reached for reading speed with the overall score, as well as with all subscale scores.

Discussion

The VA LV VFQ48 questionnaire is the most extensively validated tool to assess PROs in patients attending low-vision services and has been used in large-scale randomized clinical trials.[12]

To the best of our knowledge, no other studies have been published that report on the translation and validating of the VA LV VFQ 48 in non-English speaking countries.

We have reported on the first translation of the VA LV VFQ48 questionnaire, which we have adapted to the Italian low-vision rehabilitation setting. The VA LV VFQ48 proved to be usable with minimal adaptation, as shown by the low rate of missing values for most items. As we expected, we found a moderate correlation with all visual function measures, which was higher for reading speed.

Based on this initial assessment, we plan to develop a refined version in which specific items are rephrased or adapted to the Italian low-vision setting. Considering the participants' observations collected during the interviews, the high frequency of omitted items is due to the following: (a) the activity is not performed for age reasons: for example, some participants reported that they did not play sport because they were no longer young; (b) the activity is not performed for lack of availability: for example, some participants reported that they did not have a garden; (c) the activity is not performed for lack of interest:

for example, some participants reported that they had not written checks because they did not need to; (e) the activity is generally carried out by other people: for example, a female spouse or a support figure usually carried out domestic cleaning; and (f) the activity is not performed due to impairment or health problems: for example, some participants reported that they were unable to come down the stairs because they had other disabilities.

A limitation of this study was that we did not use Rasch analysis for the statement check and selection and to assess the psychometric properties of the questionnaire. Since this is a preliminary study, we used simple methods, such as Cronbach's alpha and correlations. We plan to use more sophisticated analytic techniques to validate a revised version of the Italian VA LV VFQ48 on a larger sample.

Based on the data collected in this study, we believe that the Italian version of the VA LV VFQ48 can be used as a structured interview in clinical practice during LVR in order to understand individual patient's needs regarding day-to-day activities and their perceived functional ability. Therefore, we recommend that further studies investigating the QOL in low-vision patients undergoing LVR in Italy use the VA LV VFQ48 as a measurement tool.

Declaration of conflicting interests

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References

1. Ehrlich JR, Spaeth GL, Carlozzi NE, et al. Patient-centered outcome measures to assess functioning in randomized controlled trials of low-vision rehabilitation: a review. *Patient* 2017; 10(1): 39–49.
2. Massof RW. A systems model for low vision rehabilitation: II. Measurement of vision disabilities. *Optom Vis Sci* 1998; 75(5): 349–373.
3. Becker SW, Lambert RW, Schulz EM, et al. An instrument to measure the activity level of the blind. *Int J Rehabil Res* 1985; 8(4): 415–424.
4. Weih LM, Hassell JB and Keeffe J. Assessment of the impact of vision impairment. *Invest Ophthalmol Vis Sci* 2002; 43: 927–935.
5. Wolffsohn JS and Cochrane AL. Design of the low vision quality-of life questionnaire (LVQOL) and measuring the outcome of low vision rehabilitation. *Am J Ophthalmol* 2000; 130(6): 793–802.
6. Stelmack JA, Szlyk JP, Stelmack TR, et al. Psychometric properties of the veterans affairs low-vision visual functioning questionnaire. *Invest Ophthalmol Vis Sci* 2004; 45(11): 3919–3928.
7. Khadka J, McAlinden C and Pesudovs K. Quality assessment of ophthalmic questionnaires: review and recommendations. *Optom Vis Sci* 2013; 90(8): 720–744.
8. Szlyk JP, Stelmack J, Massof RW, et al. Performance of the veterans affairs low vision visual functioning questionnaire. *J Vis Impair Blind* 2004; 98: 261–275.
9. Stelmack JA, Rosenbloom AA, Brenneman CS, et al. Patient's perception of need for low vision devices. *J Vis Impair Blind* 2003; 97: 521–535.
10. Stelmack JA, Szlyk JP, Stelmack TR, et al. Measuring outcomes of vision rehabilitation with the veterans affairs low vision visual functioning questionnaire. *Invest Ophthalmol Vis Sci* 2006; 47(8): 3253–3261.
11. Stelmack JA, Tang XC, Reda DJ, et al. Outcomes of the veterans affairs low vision intervention trial (LOVIT). *Arch Ophthalmol* 2008; 126: 608–617.
12. Stelmack JA, Tang XC, Reda DJ, et al. VA LOVIT II: a protocol to compare low vision rehabilitation and basic low vision. *Ophthalmic Physiol Opt* 2012; 32(6): 461–471.
13. Van de Vijver F and Tanzer NK. Bias and equivalence in cross-cultural assessment: an overview. *Eur Rev Appl Psychol* 2004; 54: 119–135.
14. Beck AT, Steer RA and Carbin MG. Psychometric properties of the beck depression inventory: twenty-five years of evaluation. *Clin Psychol Rev* 1988; 8: 77–100.
15. Diener ED, Emmons RA, Larsen RJ, et al. The satisfaction with life scale. *J Personal Assess* 1985; 49: 71–75.
16. Snyder CR, Harris C, Anderson JR, et al. The will and the ways: development and validation of an individual-differences measure of hope. *J Pers Soc Psychol* 1991; 60(4): 570–585.
17. Scheier MF, Carver CS and Bridges MW. Distinguishing optimism from neuroticism (and trait anxiety, self-mastery, and self-esteem): a reevaluation of the life orientation test. *J Pers Soc Psychol* 1994; 67(6): 1063–1078.
18. Norton PJ and Weiss BJ. The role of courage on behavioral approach in a fear-eliciting situation: a proof-of-concept pilot study. *J Anxiety Disord* 2009; 23(2): 212–217.
19. Wang C, Su S and Weiss DJ. Robustness of parameter estimation to assumptions of normality in the multidimensional graded response model. *Multivariate Behav Res* 2018; 53(3): 403–418.